PQ BMS

Powerqueen BMS Monitor

User Manual

Black Sphere Industries
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1 Introduction

PQ BMS (Powerqueen BMS Monitor) is a Progressive Web App developed by Black Sphere Industries for real-time monitoring of Powerqueen LiFePO4 battery management systems via Bluetooth Low Energy (BLE). It supports simultaneous monitoring of up to two batteries, providing state-of-charge visualization, individual cell voltage monitoring, protection status, and live history charting.

The app communicates with Powerqueen BMS units over the BLE GATT protocol using a binary command/response format, all running client-side in the browser via the Web Bluetooth API.

2 Getting Started

2.1 System Requirements

- **Browser:** Chrome, Edge, or Opera on desktop or Android. Web Bluetooth is required and is **not** supported in Firefox or Safari.
- **Bluetooth:** A BLE-capable Bluetooth adapter on the host device
- **Battery:** One or two Powerqueen LiFePO4 batteries with BLE-enabled BMS

2.2 Connecting a Battery

1. Open the PQ BMS application URL
2. Click **Connect Battery 1** on the connect screen
3. In the browser's Bluetooth pairing dialog, select your Powerqueen BMS device (filtered by service UUID 0xFFE0)
4. The dashboard appears with live battery data

2.3 Connecting a Second Battery

After the first battery is connected:

1. Click **Add Battery 2** in the toolbar
2. Pair the second BMS device
3. A dual-panel layout appears with side-by-side monitoring
4. A combined system summary banner shows aggregate statistics

3 Features

3.1 State of Charge (SOC) Display

Each battery panel features a prominent SOC ring visualization:

- A circular gauge with animated fill showing the current SOC percentage
- Color-coded: green (healthy), yellow (moderate), red (low)
- Status text below the ring indicating charge state (Charging, Discharging, Idle)

3.2 Quick Statistics

Three stat cards per battery show at-a-glance values:

- **Voltage:** Total pack voltage
- **Current:** Charge/discharge current with direction indicator
- **Power:** Instantaneous power in watts

3.3 Cell Voltage Grid

An expandable section displays individual cell voltages in a 2-column grid:

- Each cell shows its voltage with precision formatting

- A color-coded bar below each cell indicates relative voltage within the pack
- Cell imbalance is immediately visible from the bar heights

3.4 Battery Information

A collapsible section shows detailed BMS parameters:

- Total capacity and remaining capacity
- Cycle count
- Temperature readings from BMS sensors
- BMS firmware version (retrieved via a separate version command)

3.5 Protection Status

The BMS reports active protection flags. The dashboard displays these as color-coded badges:

- Green badges for normal states
- Yellow badges for warnings
- Red badges for active protection conditions (over-voltage, under-voltage, over-current, over-temperature, etc.)

3.6 Combined System Summary

When two batteries are connected, a summary banner appears above the dual-panel layout showing:

- **Total Voltage:** Sum of both pack voltages
- **Total Capacity:** Combined remaining capacity
- **Total Power:** Combined instantaneous power
- **Stored Energy:** Combined energy in watt-hours

3.7 Live History Chart

A shared chart section at the bottom of the dashboard renders rolling time-series data:

- SOC and voltage traces for each connected battery
- Color-coded per battery (amber for Battery 1, gold for Battery 2)
- Up to 60 data points of history

3.8 Auto-Refresh

The toolbar provides manual and automatic refresh options:

- **Refresh All:** Manually polls all connected BMS units
- **Auto: On/Off:** Toggles periodic automatic polling

4 User Interface

4.1 Header

The sticky header contains:

- Battery icon and "PQ BMS" title
- Connection status dots for each device (green = connected, yellow = connecting, red = disconnected)
- Device name labels

4.2 Connect Screen

Displayed when no devices are connected. Shows a floating battery animation, description text, and the **Connect Battery 1** button. A warning message appears if Web Bluetooth is not supported.

4.3 Dashboard

The main dashboard layout consists of:

- **Toolbar:** Refresh All, Auto toggle, and Add Battery 2 buttons
- **Combined Banner:** Aggregate stats (visible with 2 batteries)
- **Dual Layout:** Side-by-side battery panels
- **Each Panel:** SOC ring, quick stats, cell voltage grid, battery info, protection status
- **History Section:** Live chart with legend
- **Last Update:** Timestamp of the most recent data fetch

4.4 Toast Notifications

Error toasts appear as a red banner at the bottom center of the screen.

4.5 Theme Support

PQ BMS supports dark (default) and light themes:

- When embedded in the BSI portal, the theme is controlled by the parent frame
- In standalone mode, theme is read from localStorage
- The BSI Theme Bridge ensures consistent appearance with the portal

5 Workflows

5.1 Single Battery Monitoring

1. Connect a Powerqueen battery via the connect screen
2. Review the SOC ring for overall charge status
3. Check quick stats for voltage, current, and power
4. Expand the cell voltage section to inspect cell balance
5. Enable auto-refresh for continuous monitoring

5.2 Dual Battery System Monitoring

1. Connect Battery 1
2. Click **Add Battery 2** and connect the second unit
3. Monitor the combined system banner for total voltage, capacity, and power
4. Compare cell voltages between batteries using the side-by-side layout
5. Use the history chart to track SOC trends over time for both units

5.3 Diagnosing Cell Imbalance

1. Expand the cell voltage grid for the battery of interest
2. Compare bar heights — cells with significantly lower bars indicate imbalance
3. Note the maximum and minimum cell voltages
4. A large spread between cells may indicate the need for a balancing cycle

5.4 Checking Protection Status

1. Scroll to the protection status section in each battery panel
2. Green badges indicate normal operation
3. Yellow or red badges indicate active protection conditions
4. Address root causes (e.g., reduce load for over-current, cool the battery for over-temperature)

6 Architecture

This section presents the software architecture of PQ BMS using UML diagrams.

6.1 Architecture Overview

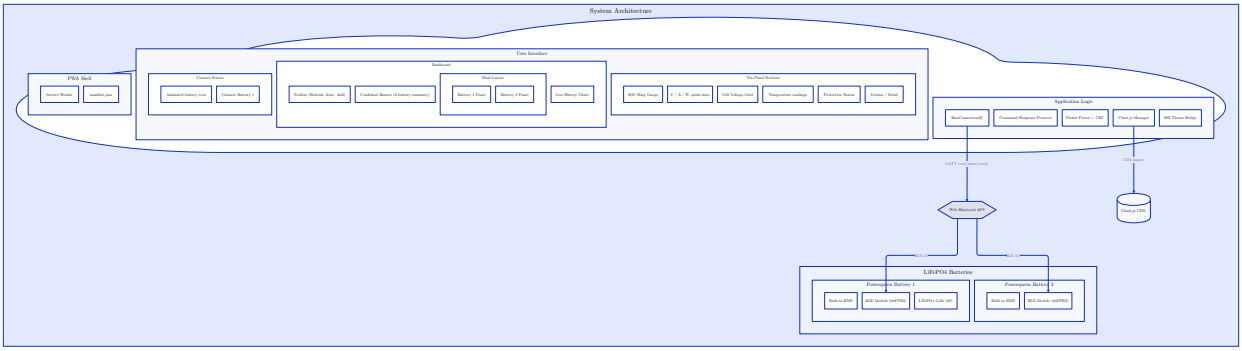

Figure 1: Component architecture of the PQ BMS application.

6.2 Class Diagram

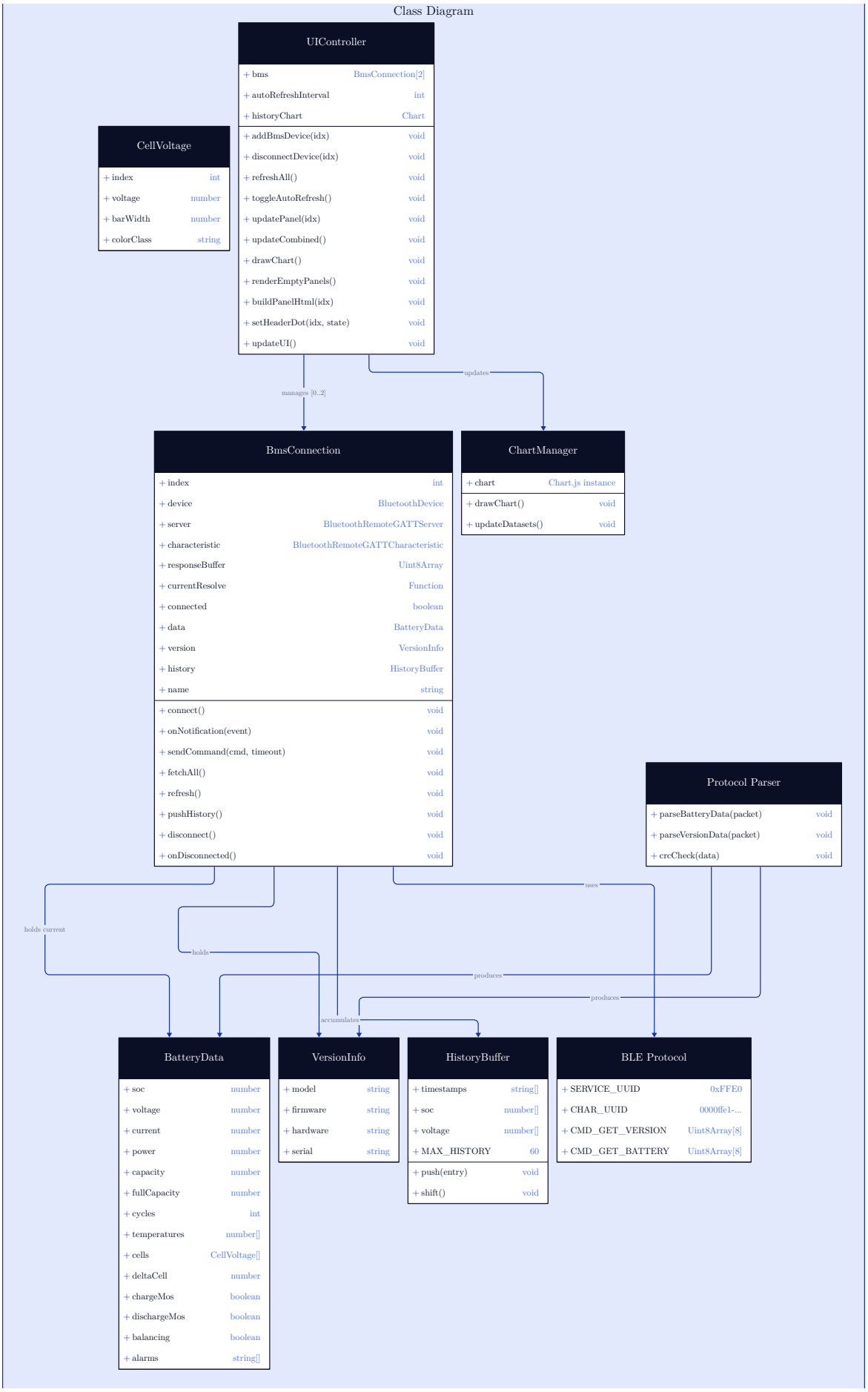

Figure 2: Class diagram showing the BmsConnection class and data structures.

6.3 Main Sequence

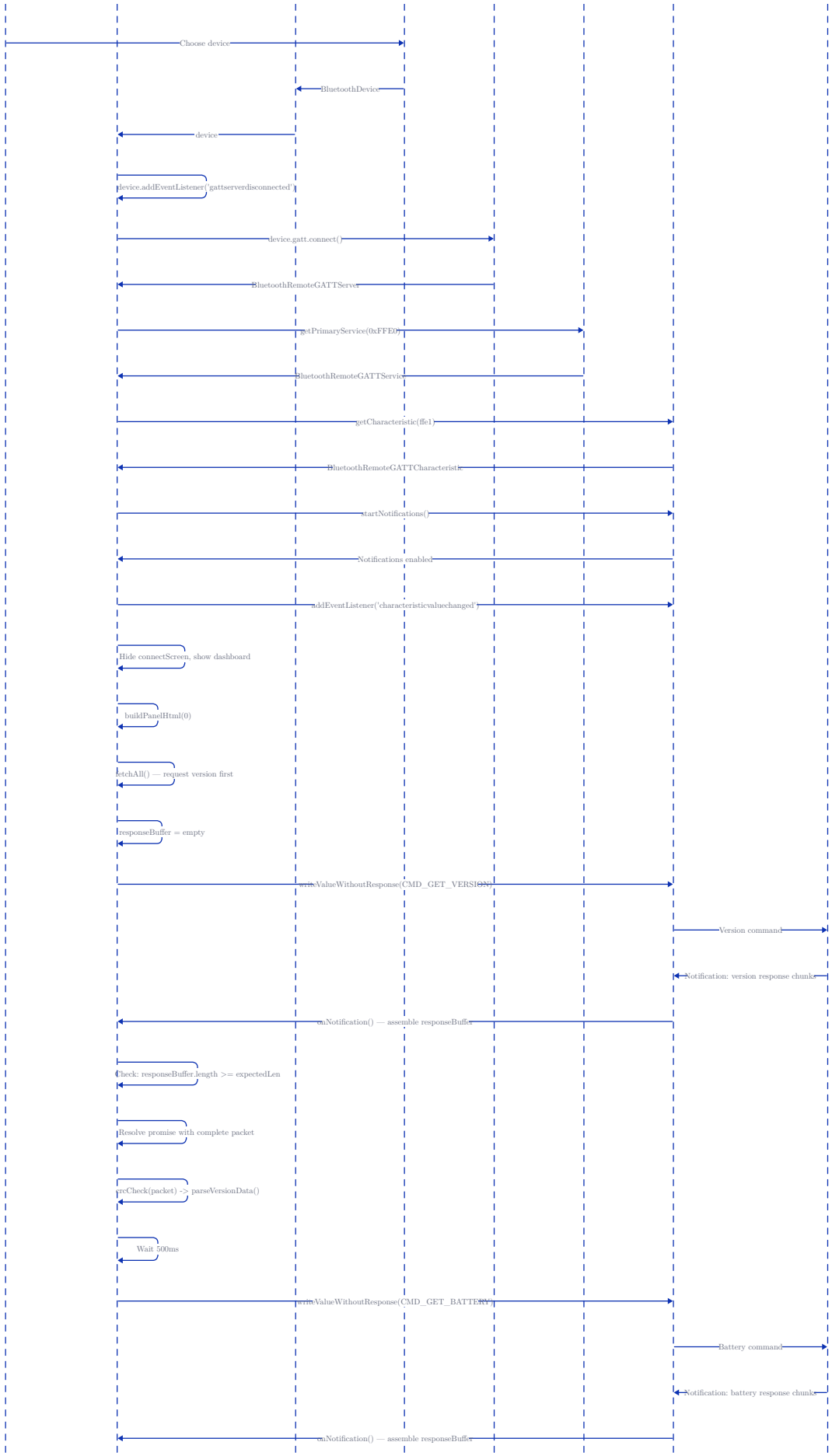

Figure 3: Sequence diagram for BLE connection and data retrieval.

6.4 Secondary Sequences

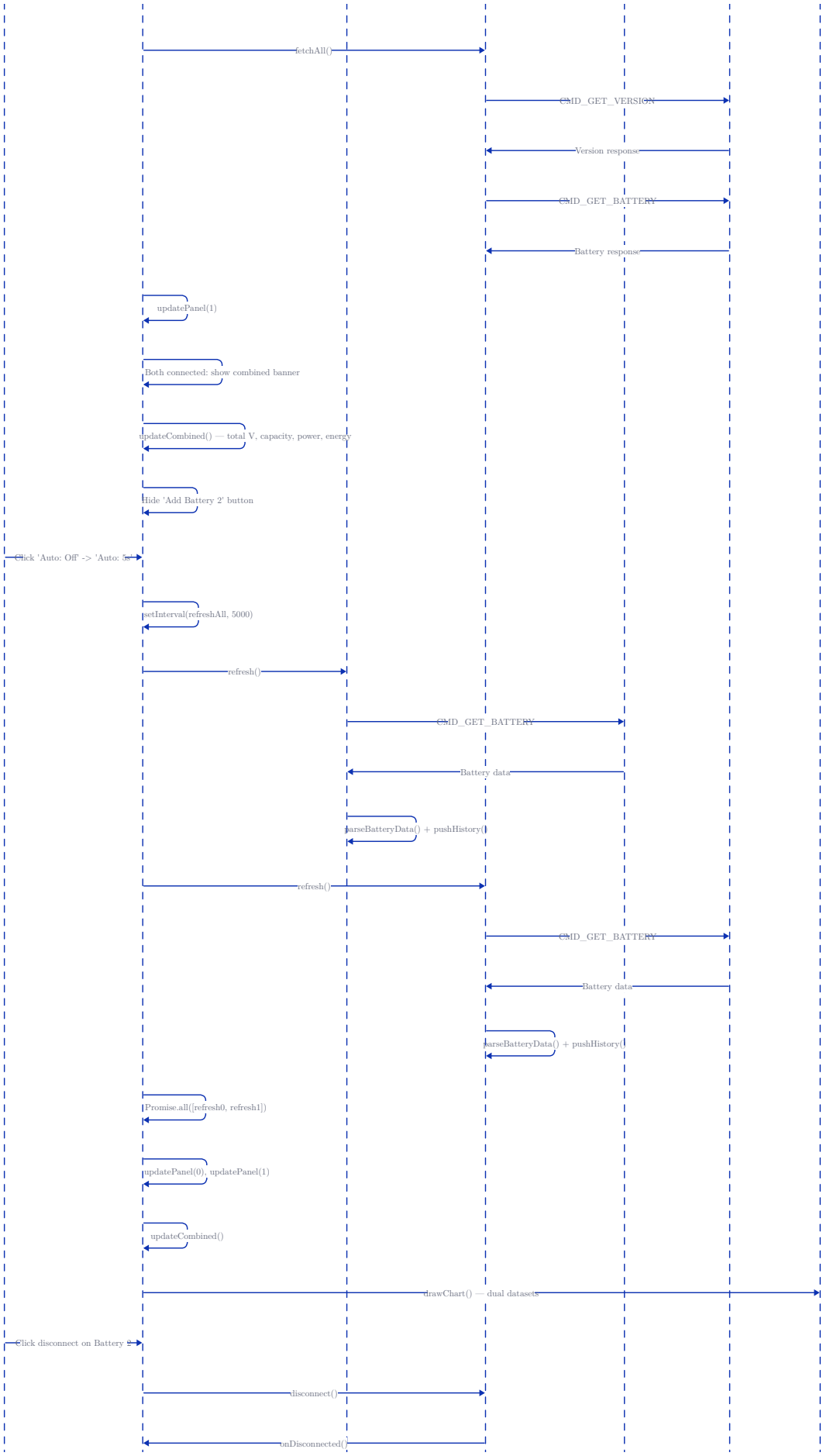

Figure 4: Sequence diagrams for auto-refresh, dual-battery, and error handling.

6.5 State Diagram

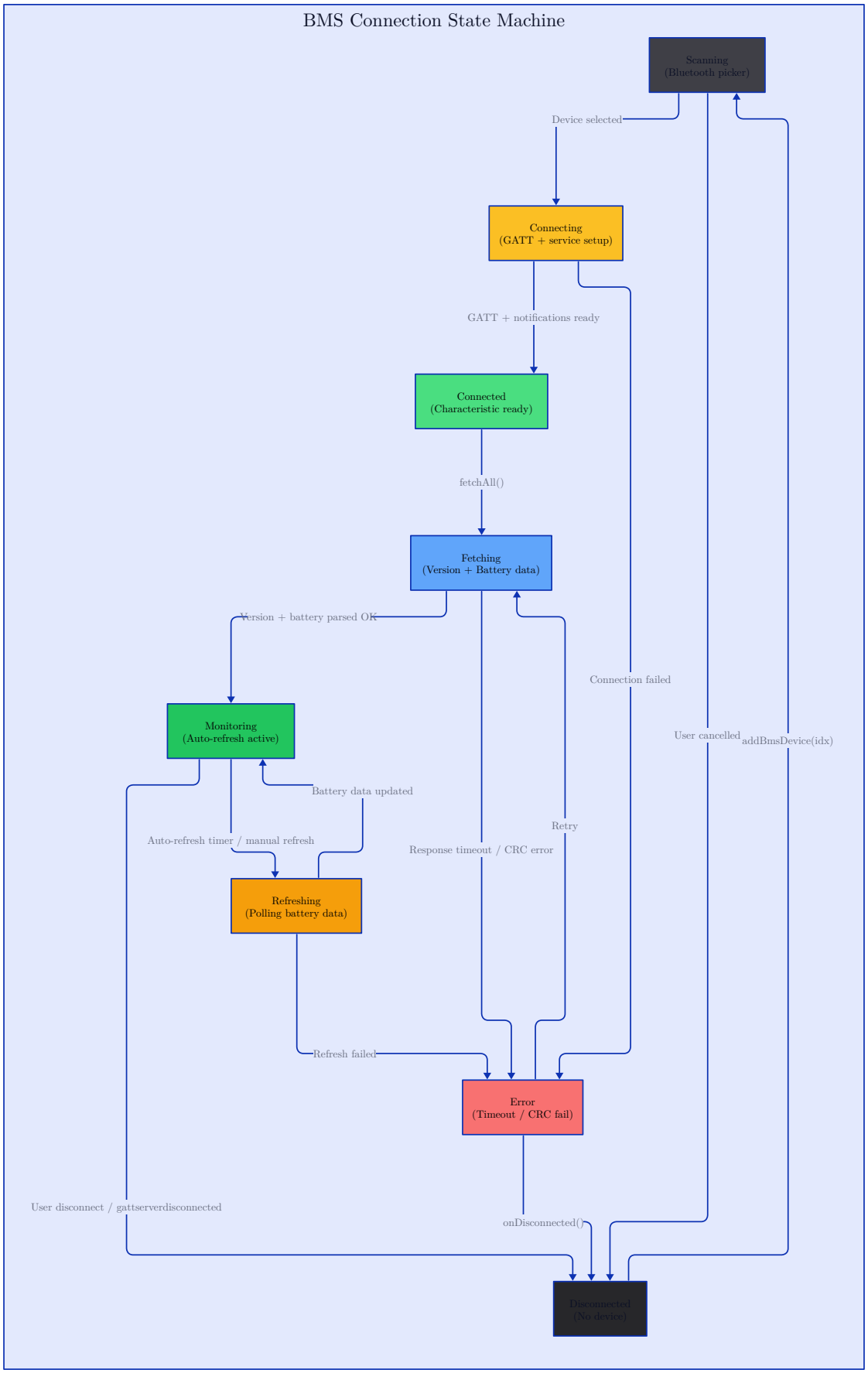

Figure 5: State machine diagram showing BLE connection states.

7 Configuration

7.1 BLE Protocol

PQ BMS communicates with the Powerqueen BMS over BLE using:

- **Service UUID:** `0xFFE0`
- **Characteristic UUID:** `0000ffe1-0000-1000-8000-00805f9b34fb`
- **Commands:** Binary command packets with header `0x55AA` and command-specific payloads
  - `CMD_GET_VERSION` — Retrieves BMS firmware version
  - `CMD_GET_BATTERY` — Retrieves full battery status (SOC, voltage, current, cell voltages, temperatures, protection flags)

7.2 Data Format

Battery data is received as a binary response buffer and parsed into:

- Pack voltage and current (signed, for charge/discharge direction)
- SOC percentage
- Individual cell voltages (variable count depending on battery configuration)
- Temperature sensor readings
- Active protection bit flags

7.3 PWA Installation

PQ BMS can be installed as a PWA via the browser's install prompt. The service worker caches all assets for offline launch (BLE connectivity still requires the browser's Bluetooth stack).

7.4 Theme Configuration

- localStorage key `pqbms_theme` stores the theme preference
- When embedded in the BSI portal, theme is controlled via postMessage

8 Troubleshooting

8.1 Common Issues

BMS device not found Ensure the Powerqueen battery is powered on and the BMS board's BLE module is active. The BLE filter requires service UUID `0xFFE0` — non-Powerqueen BMS devices will not appear.

"Web Bluetooth not supported" warning Use Chrome, Edge, or Opera. Firefox and Safari do not support Web Bluetooth.

Data shows "–" after connecting The initial data fetch may take a few seconds. Click **Refresh All** if data does not populate. Check the browser console for BLE communication errors.

Connection drops frequently Ensure the device is within BLE range (10 m line-of-sight). Interference from other Bluetooth devices or Wi-Fi can degrade BLE stability.

Cell voltages all show 0 V The BMS response parsing may not match the specific battery model's cell count. Check the browser console for protocol parsing errors.

Second battery panel is empty Click **Add Battery 2** and complete the BLE pairing dialog. Each battery requires a separate BLE connection.

SOC ring shows incorrect color SOC color thresholds are hard-coded — green above 50%, yellow 20–50%, red below 20%. These reflect typical LiFePO4 operating ranges.